

Functional Features

Project Title: Rising Waters

Introduction

The Rising Waters system is designed to predict flood risk using Machine Learning. The application provides a simple interface for users to enter weather parameters and receive instant prediction results. The system focuses on accuracy, usability, and quick response.

Functional Features

User Input

The system allows users to enter weather-related parameters such as temperature, humidity, cloud cover, annual rainfall, seasonal rainfall, average June rainfall, and subdivision rainfall through a web form.

Data Processing

The entered values are validated and preprocessed using the trained scaler before being passed to the Machine Learning model for prediction.

Flood Prediction

The trained XGBoost model analyzes the processed data and predicts whether flood conditions are likely to occur.

Probability Calculation

The application calculates and displays the probability of flood occurrence, helping users understand the level of risk.

Result Display

Based on the prediction, the system displays either a **Flood Risk** page or a **No Flood Risk** page along with the prediction probability and safety information.

Error Handling

The application validates user input and displays appropriate error messages if invalid or incomplete values are entered.

Conclusion

The functional features of the Rising Waters system ensure accurate prediction, quick processing, and an easy-to-use interface. These features help users obtain reliable flood risk information and support better disaster preparedness.